

and perfect capital mobility. It brings about currency depreciation that improves the trade balance, so that output increases from Y_0 all the way to Y_3 .

Check Your Progress 1

1. What is the effect of monetary expansion on output and interest rates in an open economy with perfect capital mobility under flexible exchange rate?

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2. Comment on the effectiveness of monetary policy in open economies in light of your answer to Question 1 above.

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3. What is the effect of fiscal contraction (cut in tax rate) on output and interest rates in an open economy with perfect capital mobility under flexible exchange rates?

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5.4 MONETARY AND FISCAL POLICIES UNDER FIXED EXCHANGE RATE

With fixed exchange rates, exchange rate fluctuations are ruled out and the central bank plays a very important role as it holds foreign exchange reserves and intervenes in foreign exchange markets to maintain the fixed parity.

5.4.1 Fiscal Policy under Fixed Exchange Rates

We first consider the impact of expansionary fiscal policy under a fixed exchange rate system, where $e = e^*$ is the official parity (Fig. 5.4).

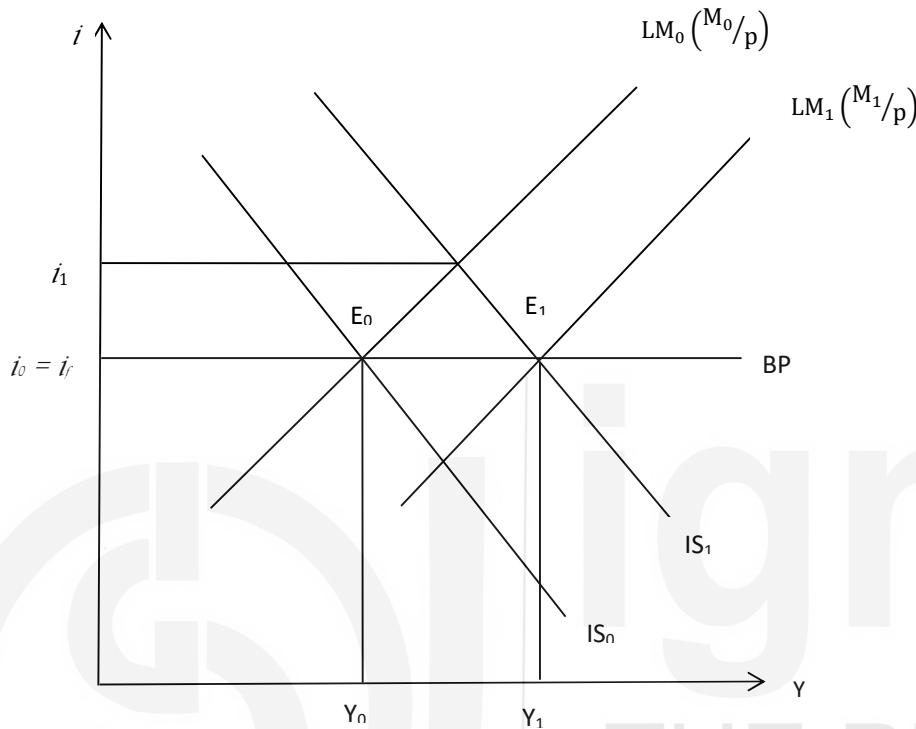


Fig. 5.4: Expansionary Fiscal Policy under Fixed Exchange Rates

Suppose there is fiscal expansion, either through an increase in government expenditure or via a cut in taxes. The IS curve shifts out from IS_0 to IS_1 and the interest rate rises to i_1 (see Fig. 5.4). With higher domestic interest rates ($i_1 > i_f$), there is a surge in capital inflows as there is BOP surplus and an excess supply of dollars (foreign exchange). This creates pressure for currency appreciation, but the central bank is committed to maintaining a fixed rate. Therefore it intervenes and buys dollars in order to mop up the excess supply. As a result foreign exchange reserves increase, leading to an increase in money supply, which shifts the LM curve outward. This process continues as long as $i > i_f$ and it stops only when LM shifts to LM_1 , where money supply has increased to M_1 ($M_1 > M_0$) and the economy reaches E_1 where $i = i_f$, output has increased to Y_1 and there is no crowding out.

Therefore, fiscal policy is extremely effective under a system of fixed exchange rates, as change in foreign exchange reserves brings about monetary expansion to accommodate the fiscal expansion. This is in complete contrast to the case of flexible exchange rates under which fiscal policy was quite ineffective.

5.4.2 Monetary Policy under Fixed Exchange Rates

Now we consider the case of a monetary expansion under a system of fixed exchange rates (Fig. 5.5).

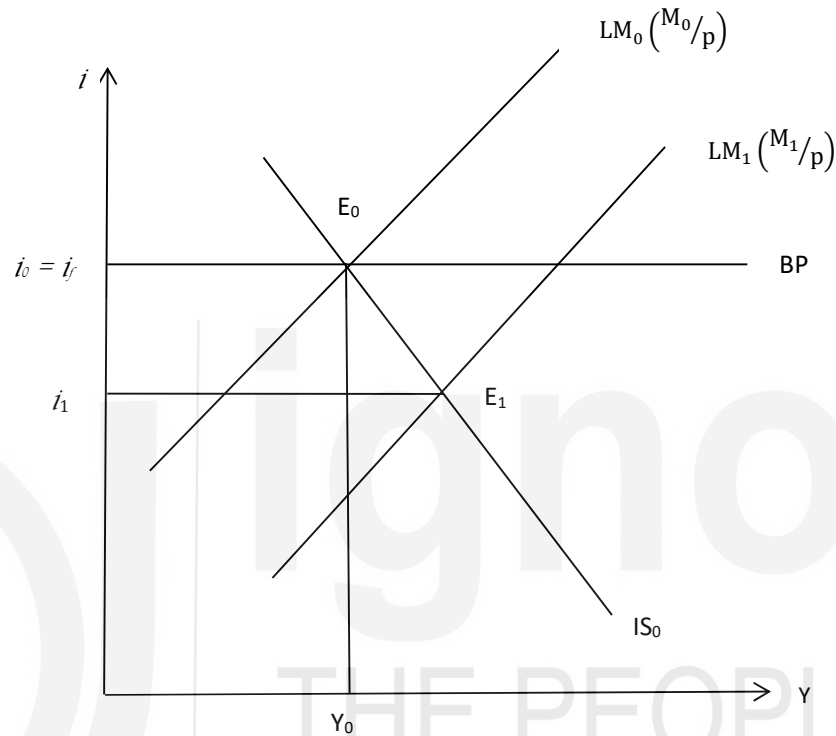


Fig. 5.5: Expansionary Monetary Policy under Fixed Exchange Rates

Suppose the central bank engages in an open market purchase of bonds in a bid to increase domestic money supply. This shifts the LM curve from LM_0 to LM_1 (since money supply increases from M_0 to M_1) and the economy moves to E_1 , where the interest rate $i_1 < i_f$ (see Fig. 5.5). Due to lower domestic rates of interest, investors would move to foreign bonds, triggering capital outflows and giving rise to a BOP deficit and excess demand for dollars. This creates pressure for currency depreciation, but the central bank is committed to maintain a fixed parity. Therefore it intervenes to sell dollars against rupees, leading to a fall in foreign exchange reserves and contraction in money supply. As money supply falls, the LM curve shifts to the left of LM_1 and this process continues as long as $i < i_f$. Ultimately the economy is back to E_0 , with money supply falling back to its initial level (M_0), so that $i = i_f$ and output is back to its initial level Y_0 .

Clearly with fixed exchange rates and perfect capital mobility, money supply becomes ineffective as a policy tool as monetary expansion has no effect on output in the short run.